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EXPERIMENT 8

Aim: Implement named entity recognizer for the given input.

Theory:

Named Entity Recognition (NER) is a subtask of Natural Language Processing (NLP) that focuses on identifying and classifying key entities within a text into predefined categories such as names of people, organizations, locations, dates, and more. By parsing through unstructured text, NER systems can extract valuable information that can be used for various applications, including information retrieval, sentiment analysis, and knowledge graph construction. The effectiveness of NER relies on a combination of linguistic rules and machine learning techniques, with modern approaches often leveraging deep learning models to improve accuracy and adaptability across different contexts and languages.

The implementation of NER has seen significant advancements with the advent of transformer-based models like BERT and GPT, which have enhanced the ability to understand context and nuances in language. These models can be fine-tuned on specific datasets to improve their performance in niche domains, such as legal or medical texts, where the types of entities may differ from general usage. As a result, NER has become an essential tool in automating data extraction processes, enabling businesses and researchers to analyze large volumes of text efficiently and derive actionable insights.

Code:



SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



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```
[1] import pandas as pd
import spacy
import requests
from bs4 import BeautifulSoup

nlp = spacy.load("en_core_web_sm")

pd.set_option("display.max_rows", 200)
```

```
[2] doc = nlp("Apple is looking at buying U.K. startup for $1 billion")

for ent in doc.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

Apple 0 5 ORG
U.K. 27 31 GPE
\$1 billion 44 54 MONEY

```
[3] from spacy import displacy

displacy.render(doc, style="ent")
```

Apple ORG is looking at buying U.K. GPE startup for \$1 billion MONEY

```
[4] entities = [(ent.text, ent.label_, ent.lemma_) for ent in doc.ents]
df = pd.DataFrame(entities, columns=['text', 'type', 'lemma'])
df
```

	text	type	lemma
0	Apple	ORG	Apple
1	U.K.	GPE	U.K.
2	\$1 billion	MONEY	\$1 billion

```
[5] content = """The owner of a Virginia company that provides the U.S. Navy with ballistic vests, protective helmets and riot gear is facing a federal wire fraud charge, accused of misleading authorities about where the products were made. Prosecutors said Arthur Morgan, the 67-year-old chief executive of Surveillance Equipment Group Inc. and its division SEG Armor, falsely claimed the equipment was made in Hong Kong and the United States when it in fact was made in China. According to federal court records, Morgan's company was an authorized seller of law enforcement and security supplies to federal agencies. Such sales must comply with the Trade Agreements Act, which requires products to be made in the United States. The United States includes Hong Kong, a special administrative region of China, on its list of countries designated to make equipment under the act but excludes the mainland. If a contractor wants to supply products from non-designated countries, "A contractor's failure to do so disqualifies the contractor from eligibility for the contract," a federal affidavit said, "and a contractor who falsely certifies cannot lawfully seek payment from the United States." Surveillance Equipment Group became an authorized equipment provider in 2003 and has fulfilled multiple federal orders over the past 16 years, the U.S. attorney for the District of Maryland said. In 2014, prosecutors said, the company was awarded a contract by the U.S. Army when a General Services Administration contracting officer asked Morgan whether his items complied with the trade act, prosecutors said, Morgan emailed saying his products were made in the "United States/Hong Kong." In 2017, he said, a federal prosecutor visited the Louisa address, on Mount Airy Road, which property records listed as a home. "Specifically, no manufacturing facility was observed at the address, which appeared, from the point of my observation, to be a residence." An Army special agent then checked pictures Morgan's firm had filed with the GSA and detected that a photo of a helmet for sale had been altered, court records said. When the agent did a reverse image search, he found the unaltered photo. Federal prosecutors researched shipping records from U.S. Customs and Border Protection and found that Morgan's company received 14 shipments of vests or helmets between 2015 and 2017 from the same Chinese firm that made the equipment. Investigators also found serial numbers on equipment that traced back to the Chinese company and Mandarin handwriting on ballistic material, prosecutors said. Emails Morgan had sent, however, indicated that orders were made in the United States. Between 2015 and July of this year, prosecutors said, five federal agencies placed nine orders for ballistic vests, helmets or riot gear from Surveillance Equipment Group, totaling about $648,000. On Thursday, a magistrate at U.S. District Court in Greenbelt, Md., ordered that Morgan be released to home confinement after he paid a $75,000 bond. An order of detention filed Friday showed he hadn't found a suitable custodian and an email and phone message left for Morgan on Sunday were not returned, and no lawyer was listed in court records. If convicted, prosecutors said, Morgan could face a maximum sentence of 20 years in federal prison. Actual sentences are typically less than that, they said. ---
```

```
[6] doc = nlp(content)
```



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0s

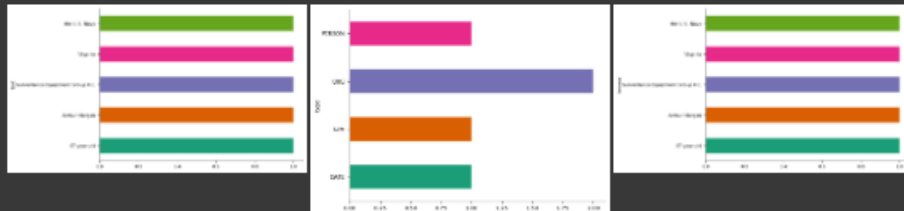
```
[7] entities = [(ent.text, ent.label_, ent.lemma_) for ent in doc.ents]
df = pd.DataFrame(entities, columns=['text', 'type', 'lemma'])
df.head()
```



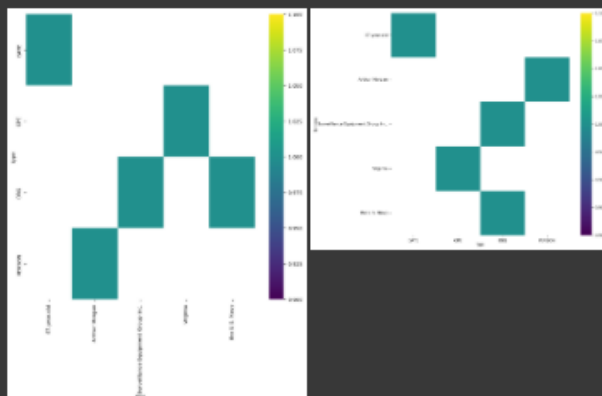
	text	type	lemma
0	Virginia	GPE	Virginia
1	the U.S. Navy	ORG	the U.S. Navy
2	Arthur Morgan	PERSON	Arthur Morgan
3	67-year-old	DATE	67-year-old
4	Surveillance Equipment Group Inc.	ORG	Surveillance Equipment Group Inc.



Categorical distributions



2-d categorical distributions





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```
df[df.type == 'GPE'].lemma.value_counts()
```

lemma	count
the United States	4
Hong Kong	3
Virginia	2
China	2
U.S.	1
the District of Maryland	1
the "United States/Hong Kong	1
Louisa	1
Va.	1
Greenbelt	1
Md.	1
not	1

dtype: int64

```
[12] # Download and parse the article
url = "https://www.revealnews.org/article/federal-judges-rulings-favored-companies-in-which-he-owned-stock/"
response = requests.get(url)
bs_doc = BeautifulSoup(response.text, 'lxml')
```

```
[20] # Pull out the article content and look at
content = bs_doc.select_one("#post-16812").text
doc = nlp(content)
```

```
[21] entities = [(ent.text, ent.label_, ent.lemma_) for ent in doc.ents]
df = pd.DataFrame(entities, columns=['text', 'type', 'lemma'])
df.head()
```

	text	type	lemma
0	Manuel Real	PERSON	Manuel Real
1	the U.S. District Court	ORG	the U.S. District Court
2	Los Angeles	GPE	Los Angeles
3	1966.Photo	CARDINAL	1966.photo
4	Virginia Lee Hunter	PERSON	Virginia Lee Hunter



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[1] [display-order \(v. 4232e-ws1-c\)](#)

Measures are in place to prevent judgments from violating federal conflict-of-interest laws. But Judge [Marcelo Real PERROW](#), 40-year [DATE](#) veteran of the bench appointed by President [Lyndon B. Johnson PERROW](#), appears to have skirted those safeguards, records and interviews show. Judges are expected to disclose judgments from their investments to their attendance at expenses paid vendors. When a financial conflict arises, no matter how small, they are required to step aside, by federal law and the [Code of Conduct for United States Law Judges](#).

"This was not what I taught for," says, meaning that it goes clear and unambiguous guidance to judges and the public," said [Steven Leifer PERROW](#), [Northwestern University PERROW](#), law professor who specializes in judicial ethics.

But in [1980s PERROW](#), cases leaving the federal [District Court LOS](#) for the [Central District LOS](#) in [Los Angeles PERROW](#). Real did not recuse himself.

[2008 PERROW](#), he awarded [Bernstein Co. PERROW](#) \$746,527 [MONEY](#) in damages and fees in a copyright infringement case against a computer sales and repair company. At the time, Real had [Microsoft PERROW](#) stock worth [between \\$11,001 and \\$50,000 PERROW](#), according to his financial disclosures in [2005 PERROW](#), case involving a contract dispute between [Atlanta Cable News PERROW](#) and [bushnell group PERROW](#). Real dismissed the suit against [Angelo PERROW](#). Real had [between \\$11,001 and \\$50,000 PERROW](#) [Angelo PERROW](#) stock, which he transferred to a company he owned as the case was appealed. The following year, Real dismissed a lawsuit against [Financial Services PERROW](#). After the plaintiff in the case appealed the decision to the [9th PERROW](#), [U.S. Circuit Court of Appeals PERROW](#), Real bought [Norton Group PERROW](#) stock worth [\\$15,000 PERROW](#) to [\\$20,000 PERROW](#). The appeals court upheld Real's decision, but returned the case to Real for further deliberation. The parties reached an agreement in [February 2011 PERROW](#), and Real dismissed the case.

In all [three PERROW](#) cases, the company's stock rose [at least PERROW](#), a dollar per share during the two-month [DATE](#) following Real's ruling or dismissal.

"When there is money involved, it is human nature to protect your own interests," said [John Schneider PERROW](#), a plaintiff in the [Verizon LOS](#) case and a retired electrical contractor. "I would say he looked out for his financial interests before he looked out for mine. Judges should be above reproach."

Real, previously recognized for your conduct on the bench, did not respond to repeated interview requests made via email or messages left on his courtroom clerk.

There is no indication that Real had a financial interest in making his decisions, and many factors affect stock prices. But judges are consulted by [California Voters PERROW](#), a project of the [Center for Investigative Reporting PERROW](#), indicated that Real's rulings were, in reviews, good news for the companies. [California Voters PERROW](#) asked law professor [Laura Levin PERROW](#), who holds the David W. Burshten Chair in Ethics at [Loyola Law School PERROW](#) [Los Angeles PERROW](#), to examine the cases.

A judge who repeatedly fails to withdraw from cases can face sanctions ranging from a private reprimand to suspension from hearing cases. Beyond that, a judge can be referred to the [Congress PERROW](#) for an impeachment hearing.

"A judge is widely considered the disqualification rules, there is a precedent for saying that should be dispositive," said [Charles Day PERROW](#), an expert on judicial ethics who teaches law at [Indiana University PERROW](#). "In addition, where judges are not diligent in keeping track of their financial conflicts, where there is a pattern of noncompliance, they could also be sanctioned."

The [Day PERROW](#) acknowledged that such sanctions are rare, typically occurring only in extreme situations. One [casebook PERROW](#) noted involved a federal district judge from [New Hampshire PERROW](#), impeached and removed from office by [the U.S. Senate PERROW](#) in [2012 PERROW](#) for failing to disclose publicly held stock in a website he accepted cash and favors from lawyers and a law books store.

Federal judges' ethics law: [the Justice Council of the 9th Circuit PERROW](#) is investigate. Judge [Ray Kippner PERROW](#), chairman of the council and not return calls seeking comment. Federal judges are required to file financial disclosures with the clerk of the court services for the [U.S. District Court PERROW](#) in [September 2008 PERROW](#). They are required to use special computer software to disclose stock holdings against their courtroom districts, although compliance has been inconsistent. Facing potential sanctions, judges generally avoid public appearances in which they discuss the law or their cases. Some judges do disclose their investments according to [Central District LOS PERROW](#) of [California PERROW](#), communications company [Day Airlines PERROW](#). The district is the largest of the [94 PERROW](#), federal judicial districts, serving a population of [more than 18 million PERROW](#).

It is a bit odd to ask judges to actually use their reports," said [Bobby Davis PERROW](#), chief of staff of the [9th U.S. Circuit Court of Appeals PERROW](#). "We are not asking the judges.... We are assuming that at that point."

Public scrutiny of the financial disclosures is complicated by logistical hurdles. Whistle blowers can request judges' [annual PERROW](#) financial disclosures through the federal court administrative office in [Stockton PERROW](#). The process of obtaining these documents and cross-checking them against court rulings is cumbersome and expensive. Judges file [annual PERROW](#) financial disclosures by [May 15 PERROW](#) in the courtroom.

[Ray PERROW](#), [Ray PERROW](#), reported to be the first in [1995 PERROW](#) to own a [\\$754,000 PERROW](#) 2-year and the other federal that court judges enjoys what has effectively become a license, a benefit engrained by authors of the [Constitution PERROW](#) to protect the court's independence. A graduate of [the University of Southern California PERROW](#), Real earned a law degree from [Loyola Law School PERROW](#) before becoming assistant [U.S. PERROW](#) attorney for the [Southern District LOS PERROW](#), [California PERROW](#), and in [1964 PERROW](#), the district's [U.S. PERROW](#) attorney.

One [casebook PERROW](#) of Real's more notable decisions was in [1972 PERROW](#) when he issued an order mandating housing to [Pasadena PERROW](#) schools.

Among judicial ethics, as well as attorneys who have argued cases before him, Real is known as an iconoclast.

[On a recent day PERROW](#), Real rarely looked up from his book as a string of attorneys tested their arguments at the top of his head. [Charles PERROW](#) by his burgundy leather chair. Real commanded the cavernous courtroom with his occasional golf drink.

Conclusion: Thus we successfully implemented named entity recognizer for the given input.